
Problem 8.16. The game of ***Nim*** is played with a collection of piles of sticks. In one move, a player may remove any nonzero number of sticks from a single pile. The players alternately take turns making moves. The player who removes the very last stick loses. Say that we have a game position in Nim with k piles containing $s_1, \ldots, s_k$ sticks. Call the position ***balanced*** if each column of bits contains an even number of 1s when each of the numbers s_i is written in binary, and the binary numbers are written as rows of a matrix aligned at the low order bits. Prove the following two facts.

Part a. Starting in an unbalanced position, a single move exists that changes the position into a balanced one.

Proof. Starting in an unbalanced position, we give steps to find a single move that changes the position into a balanced one.

1. Find the left most column k with odd number of 1s. In an unbalanced position at least one s_i will have a 1 in column k. Pick one such s_i.

2. Go from column k to column 1 in selected s_i and change 1 to 0 or 0 to 1 to get a balance position.

3. Subtract the new number produced in previous step from original s_i. The result is the number of sticks that can be removed from pile s_i to get a balance position.

$\square$

Part b. Starting in a balanced position, every single move changes the position into an unbalanced one.

Proof. Starting in a balanced position, and removing some non-zero number of sticks from a pile changes the number of sticks in the pile. This new number has a different combination of 1s and 0s. So some column will loose even number of 1s. $\square$

Part c. Let $NIM = \{\langle s_1, \ldots, s_k \rangle \mid \text{each } s_i \text{ is a binary number and Player I has a winning strategy in the Nim game starting at this position}\}$. Use the preceding facts about balanced positions to show that $NIM \in \mathrm{L}$.

Proof. $\square$